Sasha Maxine Alexander

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EDUCATION

Johns Hopkins Krieger School of Arts and Sciences | Aug 2023- Dec 2024

M.S. in Regenerative and Stem Cell Technologies

University of California, Davis | JUNE 2021

B.S. in Neurobiology, Physiology, and Behavior
Concentration in Neurobiology

EXPERIENCE

Clinical Research Coordinator, Stanford Medicine | August 2021- present
Leland Stanford Junior University, School of Medicine, Department of Neurosurgery

- Serve as primary contact for research patients, sponsors, and regulatory agencies
- Over 1000+ hours with experience in neurosurgery clinics, operating rooms, and ICUs
- Determine eligibility and obtain consent from patients based on specific study protocol
- Coordinate start up through close-out across multiple drug and device research studies
- Collect and process study samples and specimens
- Experience in administering several stroke and cognitive assessments
- Communicate and interact regularly with principal investigators to ensure patient safety and accurate study conduct
- Manage research databases, created study related documents and case report forms
- Manage regulatory requirements and ensure compliance with institutional review board renewal submissions

Research Assistant under Dr. Moss Zhao Stanford Medicine | August 2023- present
Leland Stanford Junior University, School of Medicine, Department of Neurosurgery

- Specifically took FSL Oxford course under Dr. Zhao
- Proceeded to continue with brain imaging in cerebrovascular patients
- Created code for analyzing images, utilizing BIDS for management
- Learned the advanced technique of ASL and perfusion imaging
- Aided in writing and served as coordinating with him for the following grants:
    - **AHA Second Century Early Faculty Independence Award**: "Next Generation Imaging of Brain Dynamics"
    - **AHA: Career Development Award**: " Deep Learning to Predict Cerebrovascular Reserve and Neurological Outcomes in Patients at High Risk for Stroke"
    - **Stanford MCHRI Team Science and Translational Medicine Pilot Grant**: "Pediatric Prediction Models for Brain Imaging"

Research Assistant, Translational Cognitive Affective Neuroscience Lab | 2018-2021
Dr. Tyler Lesh PhD, Imaging Center, UC Davis

- Conducted research on projects utilizing both imaging (specifically *DTI* and *fMRI*) and qualitative clinical assessment for bipolar disorder, early psychosis, and schizophrenic patients
- Developed independent thesis to use Voxel Based Morphometry (*VBM*) comparing gray matter volume in the amygdala in psychotic disorder patients taking mood stabilizers versus cannabis use
    - Constructed prior to covid, but due to pandemic could not see patients
- Acquired proficiency in RedCap, analyzing neurological imaging, and shadowed clinic visits
- Experience in conducting Structured Clinical Interview for DSM-V Disorders (*SCID-5*)

Intern and Volunteer, Johns Hopkins Sibley Memorial Hospital | 2015-2017

Emergency Department, General Surgery Unit, Alzheimer's Unit, Johns Hopkins Medicine

- Developed statistical analysis through surgical data for numerous surgeons regarding their cases
- Assisted medical team staff in Emergency Department by transporting patients
- Assisted nurses during emergency operations
- Experience within the neurodegenerative department, specifically the Alzheimer's unit and focused on patient care

Intern, Capitol Hill | 2016

Congresswoman Diana Degette, U.S. House of Representatives

- Specifically chose to work under her due to her work as a leading force in stem cell legislation
- Identified key concepts in her current work regarding healthcare and climate change
- Focused on her legislative activities regarding her service as Chair of Energy Committee and Co-Chair of the following caucuses: diabetes, pro-choice, and privacy
- Communicated with other members at capitol hill including senators and other honorable congressmen
- Attended numerous hearings and congressional discussions relating to various bills and acts

LEADERSHIP

Private Pilot | 2017-present

- Certified with FAA license and regulations in under 3 months through the part 41 program
- Obtained Single Engine Land Rating
- Developed leadership through extensive situational awareness, fast decision making, critical thinking, and time management
- Gained extensive knowledge in aerodynamics, navigation, aircraft systems, and meteorology
- Compliant under strict and highly regulated flight restrictions in D.C. Area
- Communicate with fast paced environment such as many international airport's air traffic control

Advisory Board Member, Status of Women Committee | 2019-2020

UC Davis Advisory Committee, Diversity, Equity, and Inclusion

- Appointed as board member to examine issues regarding women in UC Davis and UC Davis Health
- Communicated any changes and solutions with associate executive vice chancellor, faculty, staff, and students
- Discussed the current status in the community and identify any gaps regarding policies
- Focus on possible outcomes and improvements regarding limitations women face at UC Davis

Board Member, UC Davis Mental Health Coalition | 2020-2021

- Appointed as board member to analyze statistical data regarding mental health across all University of California schools to compare with UC Davis's demographics
- Implemented regulations and changes regarding mental health
- Created numerous solutions to increase awareness and prevention techniques
- Promoted better access to mental health care across community

Board Member, Pre-Med AMSA Board| 2018-2019

Pre-Medical American Medical Student Association, UC Davis

- Appointed as board member to present and work with researchers and health care members
- Communicated with those in the medical field to discuss the transition and dedication required
- Focused on recruiting people by instilling medical related events, activities, and talks

SKILLS

Basic Science Techniques: Flow Cytometry, PCR, reprogramming Human Somatic Cells to iPS cells, generating iPSC-derived cells via differentiation and reprogramming, transcriptomics analysis of iPSC differentiation via single-cell RNA sequencing (scRNA-Seq), and proteomics analysis of iPSC differentiation via data-independent acquisition mass spectrometry (DIA-MS)

Languages: French (*fluent*), Arabic (*proficient*), Farsi (*intermediate*)

Neurological assessments: mGCS, NIHSS, GOSE, mRS, MoCA

Psychiatric assessments: SCID-5, BD-II, C-SSRS, HDRS

Clinical Research: CITI Training in IRB BioMed/GCP Research (Group 7), GCP Social and Behavioral Research Best Practices for Clinical Research (Group 10), HIPAA, Human Bloodborne Pathogens

Software: FSL, DSI Studio, Freesurfer, Python, R programing, STATA, RedCap, iMedidata, Oracle, Excel, Word

Research and Publications

Patents

Both Patent Pending and under review

High-efficiency Spin-orbit Torque Magnetoresistive Memory Device Having Spin Reflection Layers

Inventors: Gary Steinberg, Chair; Michael E Moseley, Professor; Yize Zhao, Instructor; and **Sasha Alexander**, Clinical Research Coordinator 2

Sponsor: American Heart Association

A novel method to predict treatment outcomes in neurosurgery through An artificial intelligence method developed

Inventors: Gary Steinberg, Chair; Michael E Moseley, Professor; Yize Zhao, Instructor; and **Sasha Alexander**, Clinical Research Coordinator 2

Sponsor: American Heart Association

PAPERS

Comparative Evaluation of Single- and Multi-Delay Arterial Spin Labeling MRI in Preterm Neonates

Yeva Prysiazhniuk, Sasha Alexander, Rui Duarte Armindo, Elizabeth Tong, Kristen W Yeom, Jakub Otáhal, Martin Kynčl, Michael Moseley, Jan Petr, Moss Y Zhao, Gary K Steinberg

*Manuscript selected by **NeuroImage**, Accepted on 10/9/2025, Published*

Prysiazhniuk Y, Alexander S, Duarte Armindo R, Tong E, Yeom KW, Otáhal J, Kynčl M, Moseley M, Petr J, Zhao MY, Steinberg GK. Comparative evaluation of single- and multi-delay arterial spin labeling MRI in preterm neonates. Neuroimage. 2025 Oct 9;321:121511. doi: 10.1016/j.neuroimage.2025.121511. Epub ahead of print. PMID: 41067667

Age-related changes in cerebral blood flow and arterial transit time in children: insights from single- and multi-delay ASL

Yeva Prysiazhniuk, Rui Duarte Armindo, Sasha Alexander, Michael Moseley, Kristen W. Yeom, Jakub Otáhal, Martin Kynčl, Elizabeth Tong, Jan Petr, Moss Y. Zhao & Gary K. Steinberg

*Manuscript Selected by: **Neuroradiology**, Accepted on 10/7/2025, Published*

Prysiazhniuk Y, Duarte Armindo R, Alexander S, Moseley M, Yeom KW, Otáhal J, Kynčl M, Tong E, Petr J, Zhao MY, Steinberg GK. Age-related changes in cerebral blood flow and arterial transit time in children: insights from single- and multi-delay ASL. Neuroradiology. 2025 Oct 7. doi: 10.1007/s00234-025-03794-9. Epub ahead of print. PMID: 41055711

Characterizing pre- and post-operative cerebral blood flow and transit time in pediatric moyamoya patients using multi-delay ASL and DSC MRI

Zhao, M. Y., Alexander, S., Lopez, C. A., Zhang, H., Morton, G., Armindo, R. D., Yeom, K. W., Tong, E., Soares, B. P., Lee, S., Moseley, M., & Steinberg, G. K.

*Manuscript Selected by: **Journal of cerebral blood flow and metabolism,** Accepted on 7/11/2025, Published*

Zhao MY, Alexander S, Lopez CA, Zhang H, Morton G, Armindo RD, Yeom KW, Tong E, Soares BP, Lee S, Moseley M, Steinberg GK. Characterizing pre- and post-operative cerebral blood flow and transit time in pediatric moyamoya patients using multi-delay ASL and DSC MRI. J Cereb Blood Flow Metab. 2025 Jul 15:271678X251358979. doi: 10.1177/0271678X251358979. Epub ahead of print. PMID: 40665679; PMCID: PMC12267206

Evaluating Vascular Hemodynamics in Cerebral Proliferative Angiopathy Using Advanced Arterial Spin Labeling MRI

Rui Duarte Armindo, Moss Zhao, Sasha Alexander, Elizabeth Tong, Kristen Yeom, Michael Moseley, Pedro Vilela, and Gary Steinberg

*Manuscript selected by **S:VIN Journal**, Accepted on 11/4/2024, Published*

No PMID; https://doi.org/10.1161/SVIN.124.001552 iSSN: 2694-5746

ABSTRACTS/POSTERS

Feasibility and hemodynamic insights from singleand multi-delay ASL MRI in preterm neonates

Yeva Prysiazhniuk, Sasha Alexander, Rui Duarte Armindo, Elizabeth Tong, Kristen W Yeom, Jakub Ota´hal, Martin Kync, Michael Moseley, Gary K Steinberg, Jan Petr, Moss Y Zhao

Book of Abstracts ESMRMB 2025 Online 41st Annual Scientific Meeting 8-11 October 2025. MAGMA. 2025 Sep 29. doi: 10.1007/s10334-025-01278-8. Epub ahead of print. PMID: 41023312.

Time as a biomarker: Multi-delay ASL Enhances Cerebrovascular Assessment in Pediatric Patients with Moyamoya Disease

***Sasha Alexander**, Gabriella Morton, Chris Lopez, and Moss Zhao*

Abstract and Poster Presentation Acceptance: BRAIN AND PET Conference in Korea June 2025

(Unable to present)

Imaging-Based Approach To The Pathophysiology Of Moyamoya Disease Among Diverse Ethnicities

*Shoko Hara, Michelle Cheng, Yize Zhao, **Sasha Alexander**, & Gary K Steinberg*

*abstract accepted for **ISC 2025 Conference***

International STROKE CONFERENCE- AHA STROKE JOURNAL VOLUME 56 SUPP. 1

Hemodynamic Changes in Developing Pediatric Brain Measured with Single- and Multi-delay Arterial Spin Labeling

Yeva Prysiazhniuk, Jan Petr, ***Sasha Alexander****, Rui Duarte Armindo, Jakub Otáhal, Martin Kynčl, Michael Moseley, Elizabeth Tong, Moss Y Zhao*

Book of Abstracts ESMRMB 2024 Online 40th Annual Scientific Meeting 2–5 October 2024. Magn Reson Mater Phy 37 (Suppl 1), 1–781 (2024). Https://doi.org/10.1007/s10334-024-01191-6

Mind the (contrast) gap: arterial spin labeling and AI redefine pediatric Moyamoya treatment

Sasha Alexander*, Gabriella Morton, Chris Lopez, and Moss Zhao*

Poster Presentation at ***Stanford Wu Tsai Neurosciences Institute's 11th Annual Symposium, 2024***

CLINICAL TRIAL EXPERIENCE, COORINDATOR

Title: A Phase 1/Phase 2a Safety and Tolerability Study of Intracerebral Transplantation of Neural Stem Cells (NR1) in Subjects with Chronic Ischemic Subcortical Stroke (ISS) and Early Evaluation of Potentially Useful Efficacy Parameters: Clinical Study Protocol Including Investigator Brochure Information

Sponsor: Stanford

Stanford Principal Investigator: Dr. Gary Steinberg, MD, PhD

Title: A Prospective, Multi-Center, Double-Blind, Randomized, Placebo-Controlled, Parallel-Group, Phase 3 Study to Assess the Efficacy and Safety of Clazosentan in Preventing Clinical Deterioration Due to Delayed Cerebral Ischemia (DCI), in Adult Subjects With Aneurysmal Subarachnoid Hemorrhage (aSAH)(REACT)

Sponsor: Idorsia Pharmaceuticals Ltd.

Stanford Principal Investigator: Dr. Gary Steinberg, MD, PhD

Title: A Phase 2 Randomized, Double-Blind, Placebo-Controlled Clinical Trial to Evaluate the Safety, Efficacy and Pharmacokinetics of REC-994 in the Treatment of Symptomatic Cerebral Cavernous Malformation - REC-994-201 Medical

Sponsor: Recursion Pharmaceuticals Ltd.

Stanford Principal Investigator: Dr. Gary Steinberg, MD, PhD

Title: The Neuroscience Brain Bank: Collection of Neurosurgical Tissue for Research

Sponsor: Stanford

Stanford Principal Investigator: Dr. Gary Steinberg, MD, PhD

Title: Evaluation of the Safety and Effectiveness of Bilateral Exablate Ablation of the Pallidothalamic Tract (PTT) for the Treatment of the Motor Complications of Parkinson's Disease (PD014)

Sponsor: InSightec

Stanford Principal Investigator: Dr. Vivek Buch, MD

Title: A Pivotal Clinical Trial of the Management of the Medically-Refractory Dyskinesia Symptoms or Motor Fluctuations of Advanced Idiopathic Parkinson’s Disease With Unilateral Lesioning of the Globus Pallidum Using the ExAblate Neuro System (PD006)

Sponsor: InSightec

Stanford Principal Investigator: Dr. Vivek Buch, MD

Title: A Clinical Trial to Evaluate the Safety and Efficacy of Staged Bilateral Exablate Treatment of Medication Refractory Essential Tremor (ET005)

Sponsor: InSightec

Stanford Principal Investigator: Dr. Pejman Ghanouni, MD, PhD

Title: A Continued Access Study to Evaluate the Effectiveness and Safety of Exablate Transcranial MRgFUS Thalamotomy Treatment of Medication Refractory Essential Tremor Subjects

Sponsor: InSightec

Stanford Principal Investigator: Dr. Pejman Ghanouni, MD, PhD

Title: Improving Spine Care with Real-Time Objective Patient Tracking

Sponsor: Stryker Neurovascular

Stanford Principal Investigator: Dr. Robert Dodd, MD, PhD

Title: BrainGate2: Feasibility Study of an Intracortical Neural System for Persons with Tetraplegia

Sponsor: Stanford

Stanford Principal Investigator: Dr. Jamie Henderson, MD

Title: BHV3000-202: Phase 2: A Double-Blind, Placebo Controlled, Crossover Trial of BHV-3000 (Rimegepant) for Treatment Refractory Trigeminal Neuralgia

Sponsor: Stanford

Stanford Principal Investigator: Dr. Michael Lim, MD

Title: A prospective, Multicenter study to Evaluate the Performance and Safety of the CerebroFlo EVD Catheter: C-CERFLO-001

Sponsor: Integra LifeSciences Corporation

Stanford Principal Investigator: Dr. Lawrence Shuer, MD

Title: Prospective Talosix VCF Registry

Sponsor: Medtronic

Stanford Principal Investigator: Dr. Nick Telischak, MD

Title: Improving Spine Care with Real-Time Objective Patient Tracking

Sponsor: Stanford

Stanford Principal Investigator: Dr. Corinna Clio Zygourakis, MD